

American BioCarbon® Absorbent Pellets

Technical Specification Sheet

100% PLANT BASED AND BIODEGRADABLE

5X Absorbent Pellets

American BioCarbon's renewable absorbent pellets absorb up to **5× their own weight in water**, double the capacity of wood-based pellets and roughly five times that of clay. Manufactured from single-source recycled sugarcane bagasse, they ship and store stable, then disintegrate on contact to expose the full internal pore structure.

Core Specifications

Property	Value
Absorption capacity	Up to 5:1
Bulk density	30–40 lb/ft ³
Moisture content	5–10%
Pellet diameter	6–8 mm
Pellet length	10–25 mm
Dust content	<10%
Biodegradability	100%
Materials	100% sugarcane bagasse
Chemical additives	None (native lignin binder)

Plant-Based Advantage

Made from single-source recycled sugarcane bagasse. The cane stalk carries a native vascular structure with open parenchyma cells. Natural lignin binds the pellet during pressing, with no synthetic adhesives in the product or the finished waste.

Flexible Use Applications

- Oil and gas waste solidification
- Environmental remediation
- Animal bedding
- Chemical and hydrocarbon spill response
- Landfill leachate management
- Dredged-sediment dewatering
- Slurry-wall and trench-spoil management

Powerful Sustainability

- Produced from 100% repurposed single-source organic plant material
- Diverts methane emissions from ag-waste byproduct of sugar cane (bagasse) for beneficial use
- Manufactured using renewable biomass energy
- 100% chemical-free and biodegradable
- No persistent polymers in landfill leachate
- Supports recycled-content credits and low-embodied-carbon specifications

How It Compares

Property	ABC Bagasse	Wood Pellet	Premium Clay
Absorption ratio	Up to 5:1	2.0–2.5:1	~1:1
Moisture content	5–10%	8–12%	1–5%
Dust content	<10%	10–20%	High
Biodegradable	Yes	Yes	No
Indicative bulk price	\$250–350/MT	\$250–400/ton	\$400–700/ton
Cost per gallon absorbed*	\$0.18	\$0.33–0.50	\$1.30–2.50

*Water basis, at indicative bulk pricing.

Freight Mathematics

One 44,000-lb truckload of ABC bagasse absorbs roughly 26,400 gallons of water-based liquid. The same truckload of wood pellet covers 13,200 gallons; of clay, 5,275 gallons.

One truck of bagasse = 2 trucks of wood pellet = 5 trucks of clay

Honeycomb Microstructure Holds More

Peer-reviewed studies of bagasse-derived sorbents describe a honeycomb-like internal structure with cavities and channels at the micro and macro scale (Castro et al., 2017). Native wood porosity is dominated by sub-2-nanometer pathways and slow capillary uptake (Plötze et al., 2020). Bagasse moves liquid faster, holds more of it, and breaks down on schedule.

Bulk Quantities

- Bulk shipped via truckload
- 1-ton super sacks
- 40-lb retail bags (palletized)

Contact & Ordering

Daniel Villamizar

(phone)

dvillamizar@americanbiocarbon.com

Victor Jehle

561-593-4168 ext 126

victor.jehle@americanbiocarbon.com

American BioCarbon, LLC

32525 Highway 1 South, White Castle, LA 70788

americanbiocarbon.com

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